Exploratory Data Analysis

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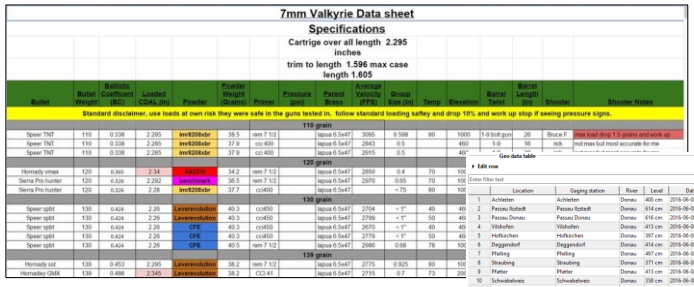
Exploratory Data Analysis

- Shorten for EDA
- Process of finding data insights by examining features/attributes/ variables.
- From features/attributes/variables, summarize main characteristics using statistics measures and visualizations.

2

Motivation

- What to do when you are given data to study?



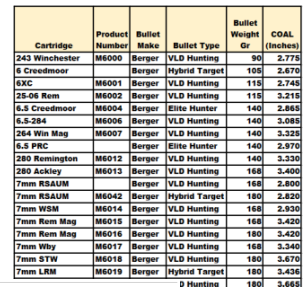
7mm Valkyrie Data sheet

Specifications

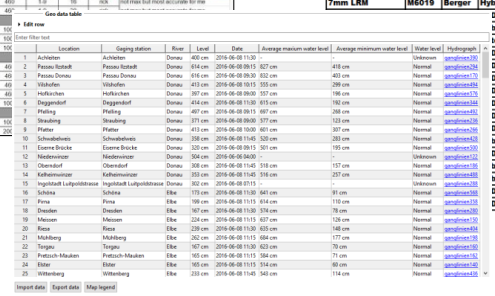
Cartrige over all length 2.295 inches

trim to length 1.596 max case length 1.605

Bullet	Bullet Weight	Ballistic Coefficent (BC)	Loaded COAL (in)	Powder	Powder Weight (Grains)	Primer	Pressure (psi)	Parent Brass	Average Velocity (FPS)	Group Size (in)	Temp	Elevation	Barrel Twist	Barrel Length (in)	Shooter	Shooter Notes
Standard disclaimer, use loads at own risk they were safe in the guns tested in. follow standard loading saftey and drop 10% and work up stop if seeing pressure signs.																
110 grain																
Speer TNT	110	0.338	2.295	inv8208xbr	38.5	rem 7 1/2		lapua 6.5x47	3095	0.598	80	1000	1-9 bolt gun	26	Bruce F	rope load drop 1.5 grains and work up
Speer TNT	110	0.338	2.285	inv8208xbr	37.9	cci 400		lapua 6.5x47	2842	0.5		460	1:8	18	nick	ind mas but most accurate for me
Speer TNT	110	0.338	2.285	inv8208xbr	37.9	cci 400		lapua 6.5x47	2915	0.5		40[illegible]				
120 grain																
Hornady vmax	120	0.365	2.34	AA2520	34.2	rem 7 1/2		lapua 6.5x47	2850	0.4	70	10[illegible]				
Sierra Pro hunter	120	0.326	2.292	Swichback	36.5	rem 7 1/2		lapua 6.5x47	2970	0.65	70	10[illegible]				
Swrra Pro hunter	120	0.326	2.28	inv8208xbr	37.7	cci400		lapua 6.5x47		<.75	80	10[illegible]				
130 grain																
Speer sptbt	130	0.424	2.26	Leverevolution	40.3	cci450		lapua 6.5x47	2704	< 1"	40	46[illegible]				
Speer sptbt	130	0.424	2.26	Leverevolution	40.3	cci450		lapua 6.5x47	2799	< 1"	50	46[illegible]				
Speer sptbt	130	0.424	2.26	CFE	40.3	cci450		lapua 6.5x47	2670	< 1"	40	46[illegible]				
Speer sptbt	130	0.424	2.26	CFE	40.3	cci450		lapua 6.5x47	2779	< 1"	50	46[illegible]				
Speer sptbt	130	0.424	2.26	CFE	40.5	rem 7 1/2		lapua 6.5x47	2986	0.68	78	10[illegible]				
139 grain																
Hornady sst	139	0.453	2.295	Leverevolution	38.2	rem 7 1/2		lapua 6.5x47	2775	0.925	80	10[illegible]				
Hornadey GMX	139	0.486	2.345	Leverevolution	38.2	CCI 41		lapua 6.5x47	2715	0.7	73	20[illegible]				



Cartridge	Product Number	Bullet Make	Bullet Type	Bullet Weight Gr	COAL (Inches)
243 Winchester	M6000	Berger	VLD Hunting	90	2.775
6 Creedmoor		Berger	Hybrid Target	105	2.670
6XC	M6001	Berger	VLD Hunting	115	2.745
25-06 Rem	M6002	Berger	VLD Hunting	115	3.215
6.5 Creedmoor	M6004	Berger	Elite Hunter	140	2.865
6.5-284	M6006	Berger	VLD Hunting	140	3.085
264 Win Mag	M6007	Berger	VLD Hunting	140	3.325
6.5 PRC		Berger	Elite Hunter	140	2.970
280 Remington	M6012	Berger	VLD Hunting	140	3.330
280 Ackley	M6013	Berger	VLD Hunting	168	3.400
7mm RSAUM		Berger	VLD Hunting	168	2.800
7mm RSAUM	M6042	Berger	Hybrid Target	180	2.820
7mm WSM	M6014	Berger	VLD Hunting	168	2.930
7mm Rem Mag	M6015	Berger	VLD Hunting	168	3.420
7mm Rem Mag	M6016	Berger	VLD Hunting	180	3.420
7mm Wby	M6017	Berger	VLD Hunting	168	3.340
7mm STW	M6018	Berger	VLD Hunting	180	3.670
7mm LRM	M6019	Berger	Hybrid Target	180	3.436
			D Hunting	180	3.665
			brid Target	180	3.610
			te Hunter	195	3.630
			D Hunting	168	2.800
			D Hunting	168	3.380
			D Hunting	190	3.510
			D Hunting	190	3.650
			D Hunting	210	3.650
			D Hunting	190	3.650
			D Hunting	190	3.650
			brid Target	215	3.555
			D Hunting	190	3.650
			brid Target	230	3.650
			brid Target	230	3.615
			D Hunting	300	3.550
			D Hunting	300	3.600



Geo data table

Edit row

Enter filter text

	Location	Gaging station	River	Level	Date	Average maxium water level	Average minimum water level	Water level	Hydrograph
1	Achleiten	Achleiten	Donau	400 cm	2016-06-08 11:30	-		Unknown	openlinex330
2	Passau Ilzstadt	Passau Ilzstadt	Donau	614 cm	2016-06-08 09:15	827 cm	418 cm	Normal	openlinex294
3	Passau Donau	Passau Donau	Donau	616 cm	2016-06-08 09:30	832 cm	403 cm	Normal	openlinex170
4	Vilshofen	Vilshofen	Donau	413 cm	2016-06-08 10:15	555 cm	299 cm	Normal	openlinex494
5	Hofkirchen	Hofkirchen	Donau	397 cm	2016-06-08 09:00	557 cm	196 cm	Normal	openlinex176
6	Deggendorf	Deggendorf	Donau	414 cm	2016-06-08 11:30	615 cm	192 cm	Normal	openlinex344
7	Pfelling	Pfelling	Donau	407 cm	2016-06-08 09:15	697 cm	268 cm	Normal	openlinex492
8	Straubing	Straubing	Donau	371 cm	2016-06-08 09:00	577 cm	123 cm	Normal	openlinex236
9	Pfatter	Pfatter	Donau	413 cm	2016-06-08 10:00	601 cm	307 cm	Normal	openlinex326
10	Schwabelweis	Schwabelweis	Donau	358 cm	2016-06-08 11:45	520 cm	283 cm	Normal	openlinex428
11	Eiserne Brücke	Eiserne Brücke	Donau	320 cm	2016-06-08 09:15	501 cm	195 cm	Normal	openlinex320
12	Niederwinzer	Niederwinzer	Donau	504 cm	2016-06-06 04:00	-	-	Unknown	openlinex122
13	Oberndorf	Oberndorf	Donau	308 cm	2016-06-08 11:45	518 cm	157 cm	Normal	openlinex186
14	Kelheimwinzer	Kelheimwinzer	Donau	353 cm	2016-06-08 11:45	516 cm	257 cm	Normal	openlinex488
15	Ingolstadt Ludwoldstrasse	Ingolstadt Ludwoldstrasse	Donau	302 cm	2016-06-08 07:15	-	-	Unknown	openlinex288
16	Schöna	Schöna	Elbe	173 cm	2016-06-08 11:30	641 cm	91 cm	Normal	openlinex588
17	Pirna	Pirna	Elbe	199 cm	2016-06-08 11:15	614 cm	110 cm	Normal	openlinex338
18	Dresden	Dresden	Elbe	167 cm	2016-06-08 11:30	574 cm	78 cm	Normal	openlinex280
19	Meissen	Meissen	Elbe	224 cm	2016-06-08 11:15	637 cm	126 cm	Normal	openlinex130
20	Riesa	Riesa	Elbe	238 cm	2016-06-08 11:30	635 cm	148 cm	Normal	openlinex684
21	Mühlberg	Mühlberg	Elbe	262 cm	2016-06-08 11:15	684 cm	177 cm	Normal	openlinex196
22	Torgau	Torgau	Elbe	167 cm	2016-06-08 11:30	623 cm	70 cm	Normal	openlinex180
23	Pretzsch-Mauken	Pretzsch-Mauken	Elbe	165 cm	2016-06-08 11:15	584 cm	71 cm	Normal	openlinex182
24	Elster	Elster	Elbe	165 cm	2016-06-08 11:15	514 cm	60 cm	Normal	openlinex140
25	Wittenberg	Wittenberg	Elbe	233 cm	2016-06-08 11:45	543 cm	114 cm	Normal	openlinex438

Import data | Export data | Map legend

Image source: https://www.7mmvalkyrie.com/load-data.html
https://www.gunwerks.com/store/featured-products/gunwerks-ammo
https://www.subshell.com/en/sophora/documentation/add-ons/data-driven-journalism-with-maps/index.html

Motivation (cont.)

- Find out as much as you can about the data

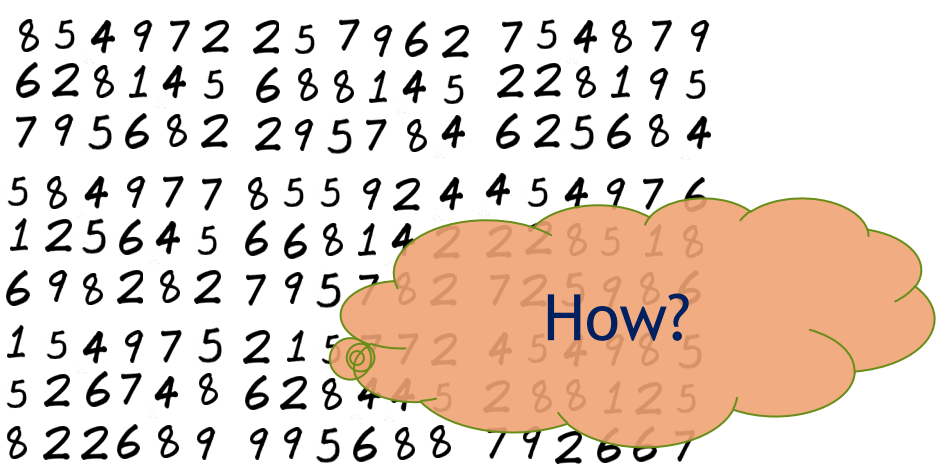


Image source: https://openclipart.org

Getting to Know Data

- Which one will you do first?

1. Find out about average and standard deviation
2. Find out members (range of values)
3. Find out distributions
4. Find out data types
5. Find out missing values
6. Find out interaction between data

Getting to Know Data

- General form of data
  - Column = feature
  - Row = object with feature values

Geo data table

Edit row

Enter filter text

	Location	Geging station	River	Level	Date	Average maxium water level	Average minimum water level	Water level	Hydrograph
1	Achleiten	Achleiten	Donau	400 cm	2016-06-08 11:30	-	-	Unknown	ganglinien390
2	Passau Ilzstadt	Passau Ilzstadt	Donau	614 cm	2016-06-08 09:15	827 cm	418 cm	Normal	ganglinien294
3	Passau Donau	Passau Donau	Donau	616 cm	2016-06-08 09:30	832 cm	403 cm	Normal	ganglinien170
4	Vilshofen	Vilshofen	Donau	413 cm	2016-06-08 10:15	555 cm	299 cm	Normal	ganglinien694
5	Hofkirchen	Hofkirchen	Donau	397 cm	2016-06-08 09:00	557 cm	196 cm	Normal	ganglinien576
6	Deggendorf	Deggendorf	Donau	414 cm	2016-06-08 11:30	615 cm	192 cm	Normal	ganglinien344
7	Pfelling	Pfelling	Donau	497 cm	2016-06-08 09:15	697 cm	268 cm	Normal	ganglinien492
8	Straubing	Straubing	Donau	371 cm	2016-06-08 09:00	577 cm	123 cm	Normal	ganglinien236
9	Pfatter	Pfatter	Donau	413 cm	2016-06-08 10:00	601 cm	307 cm	Normal	ganglinien266
10	Schwabelweis	Schwabelweis	Donau	358 cm	2016-06-08 11:45	520 cm	283 cm	Normal	ganglinien428
11	Eserne Brücke	Eserne Brücke	Donau	320 cm	2016-06-08 09:15	501 cm	195 cm	Normal	ganglinien500
12	Niederwinzer	Niederwinzer	Donau	504 cm	2016-06-06 04:00	-	-	Unknown	ganglinien122
13	Oberndorf	Oberndorf	Donau	308 cm	2016-06-08 11:45	518 cm	157 cm	Normal	ganglinien186
14	Kelheimwinzer	Kelheimwinzer	Donau	353 cm	2016-06-08 11:45	516 cm	257 cm	Normal	ganglinien488
15	Ingolstadt Luitpoldstrasse	Ingolstadt Luitpoldstrasse	Donau	302 cm	2016-06-08 07:15	-	-	Unknown	ganglinien288
16	Schöna	Schöna	Elbe	173 cm	2016-06-08 11:30	641 cm	91 cm	Normal	ganglinien568
17	Pirna	Pirna	Elbe	199 cm	2016-06-08 11:15	614 cm	110 cm	Normal	ganglinien358
18	Dresden	Dresden	Elbe	167 cm	2016-06-08 11:30	574 cm	78 cm	Normal	ganglinien280
19	Meissen	Meissen	Elbe	224 cm	2016-06-08 11:15	637 cm	126 cm	Normal	ganglinien130
20	Riesa	Riesa	Elbe	239 cm	2016-06-08 11:30	635 cm	148 cm	Normal	ganglinien404
21	Muhlberg	Muhlberg	Elbe	262 cm	2016-06-08 11:15	684 cm	177 cm	Normal	ganglinien198
22	Torgau	Torgau	Elbe	167 cm	2016-06-08 11:30	623 cm	70 cm	Normal	ganglinien160
23	Pretzsch-Mauken	Pretzsch-Mauken	Elbe	165 cm	2016-06-08 11:15	584 cm	71 cm	Normal	ganglinien162
24	Elster	Elster	Elbe	165 cm	2016-06-08 11:15	514 cm	60 cm	Normal	ganglinien140
25	Wittenberg	Wittenberg	Elbe	233 cm	2016-06-08 11:45	543 cm	114 cm	Normal	ganglinien436

Import data Export data Map legend

Cartridge	Product Number	Bullet Make	Bullet Type	Bullet Weight Gr	COAL (Inches)
243 Winchester	M6000	Berger	VLD Hunting	90	2.775
6 Creedmoor		Berger	Hybrid Target	105	2.670
6XC	M6001	Berger	VLD Hunting	115	2.745
25-06 Rem	M6002	Berger	VLD Hunting	115	3.215
6.5 Creedmoor	M6004	Berger	Elite Hunter	140	2.865
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6.5 PRC		Berger	Elite Hunter	140	2.970
280 Remington	M6012	Berger	VLD Hunting	140	3.330
280 Ackley	M6013	Berger	VLD Hunting	168	3.400
7mm RSAUM		Berger	VLD Hunting	168	2.800
7mm RSAUM	M6042	Berger	Hybrid Target	180	2.820
7mm WSM	M6014	Berger	VLD Hunting	168	2.930
7mm Rem Mag	M6015	Berger	VLD Hunting	168	3.420
7mm Rem Mag	M6016	Berger	VLD Hunting	180	3.420
7mm Wby	M6017	Berger	VLD Hunting	168	3.340
7mm STW	M6018	Berger	VLD Hunting	180	3.670
7mm LRM	M6019	Berger	Hybrid Target	180	3.436
7mm RUM	M6020	Berger	VLD Hunting	180	3.665
28 Nosler	M6041	Berger	Hybrid Target	180	3.610
28 Nosler	M6042	Berger	Elite Hunter	195	3.630
308 Win	M6021	Berger	VLD Hunting	168	2.800
30-06	M6022	Berger	VLD Hunting	168	3.380
300 Win Mag	M6024	Berger	VLD Hunting	190	3.510
300 Wby	M6026	Berger	VLD Hunting	190	3.650
300 RUM		Berger	VLD Hunting	210	3.650
300 RUM		Berger	VLD Hunting	190	3.650
300 RUM		Berger	VLD Hunting	190	3.650
30 Nosler	M6043	Berger	Hybrid Target	215	3.555
300 RUM	M6028	Berger	VLD Hunting	190	3.650
300 RUM	M6035	Berger	Hybrid Target	230	3.650
300 Norma	M6032	Berger	Hybrid Target	230	3.615
338 Norma	M6033	Berger	VLD Hunting	300	3.550
338 Lapua	M6034	Berger	VLD Hunting	300	3.600

Image source: https://www.gunwerks.com/store/featured-products/gunwerks-ammo
https://www.subshell.com/en/sophora/documentation/add-ons/data-driven-journalism-with-maps/index.html

Getting to Know Data

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Getting to Know Data

- Find out data types
  - What is the data type of each column?
- Find out meaning of each column

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15	Ingolstadt Luitpoldstrasse	Ingolstadt Luitpoldstrasse	Donau	302 cm	2016-06-08 07:15	-	-	Unknown	ganglinien288
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23	Pretzsch-Mauken	Pretzsch-Mauken	Elbe	165 cm	2016-06-08 11:15	584 cm	71 cm	Normal	ganglinien162
24	Elster	Elster	Elbe	165 cm	2016-06-08 11:15	514 cm	60 cm	Normal	ganglinien140
25	Wittenberg	Wittenberg	Elbe	233 cm	2016-06-08 11:45	543 cm	114 cm	Normal	ganglinien436

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28 Nosler	M6042	Berger	Elite Hunter	195	3.630
308 Win	M6021	Berger	VLD Hunting	168	2.800
30-06	M6022	Berger	VLD Hunting	168	3.380
300 Win Mag	M6024	Berger	VLD Hunting	190	3.510
300 Wby	M6026	Berger	VLD Hunting	190	3.650
300 RUM		Berger	VLD Hunting	210	3.650
300 RUM		Berger	VLD Hunting	190	3.650
300 RUM		Berger	VLD Hunting	190	3.650
30 Nosler	M6043	Berger	Hybrid Target	215	3.555
300 RUM	M6028	Berger	VLD Hunting	190	3.650
300 RUM	M6035	Berger	Hybrid Target	230	3.650
300 Norma	M6032	Berger	Hybrid Target	230	3.615
338 Norma	M6033	Berger	VLD Hunting	300	3.550
338 Lapua	M6034	Berger	VLD Hunting	300	3.600

Image source: https://www.gunwerks.com/store/featured-products/gunwerks-ammo
https://www.subshell.com/en/sophora/documentation/add-ons/data-driven-journalism-with-maps/index.html

Data Types

1. Categorical Data
   - Nominal data : cannot be sorted
   - Ordinal data
2. Numerical data
   - Interval data : ratio does not have meaning
   - Ratio data

Comparison: Type of Data

- Comparison

	Nominal	Ordinal	Interval	Ratio
Can sort values		✓	✓	✓
Can find sum/differences			✓	✓
Can find ratios				✓
Has fixed zero points				✓

Image source: https://www.mymarketresearchmethods.com/types-of-data-nominal-ordinal-interval-ratio/

Getting to Know Data

- Example
  - Which data is nominal, ordinal, interval, and ratio?

total_bill	tip	sex	smoker	day	time	size
16.99	1.01	Female	No	Sun	Dinner	2
10.34	1.66	Male	No	Sun	Dinner	3
21.01	3.5	Male	No	Sun	Dinner	3
23.68	3.31	Male	No	Sun	Dinner	2
24.59	3.61	Female	No	Sun	Dinner	4
25.29	4.71	Male	No	Sun	Dinner	4
8.77	2	Male	No	Sun	Dinner	2
26.88	3.12	Male	No	Sun	Dinner	4
15.04	1.96	Male	No	Sun	Dinner	2
14.78	3.23	Male	No	Sun	Dinner	2
10.27	1.71	Male	No	Sun	Dinner	2
35.26	5	Female	No	Sun	Dinner	4
15.42	1.57	Male	No	Sun	Dinner	2
18.43	3	Male	No	Sun	Dinner	4

Getting to Know Data

Pclass = Passenger class
SibSp = # of Sibling or Spouse
Parch = # of Parent or Children

Ticket = Ticket number
Cabin = Cabin number
Embarked = Embarking Part
(C = Cherbourg,
Q = Queenstown,
S = Southampton)

- Example 2
  - Which data is nominal, ordinal, interval, and ratio?

Passenger	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25		S
2	1	1	Cumings, Mrs. John Brad	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2.	7.925		S
4	1	1	Futrelle, Mrs. Jacques He	female	35	1	0	113803	53.1	C123	S
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.05		S
6	0	3	Moran, Mr. James	male		0	0	330877	8.4583		Q
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51.8625	E46	S
8	0	3	Palsson, Master. Gosta L	male	2	3	1	349909	21.075		S
9	1	3	Johnson, Mrs. Oscar W (E	female	27	0	2	347742	11.1333		S
10	1	2	Nasser, Mrs. Nicholas (A	female	14	1	0	237736	30.0708		C
11	1	3	Sandstrom, Miss. Margu	female	4	1	1	PP 9549	16.7	G6	S
12	1	1	Bonnell, Miss. Elizabeth	female	58	0	0	113783	26.55	C103	S
13	0	3	Saundercock, Mr. Williar	male	20	0	0	A/5. 2151	8.05		S
14	0	3	Andersson, Mr. Anders J	male	39	1	5	347082	31.275		S
15	0	3	Vestrom, Miss. Hulda An	female	14	0	0	350406	7.8542		S
16	1	2	Hewlett, Mrs. (Mary D Ki	female	55	0	0	248706	16		S

Getting to Know Data

- Which one will you do first?

1. Find out about average and standard deviation
2. Find out members (range of values)
3. Find out distributions
4. Find out data types
5. Find out missing values
6. Find out interaction between data

Exercise

- How to deal with null values?
  - Keep or Drop
    - Avoid dropping (if majority of data are missing)
    - Drop by row or by column
  - Keep -> Replace with representative value
    - Generally representative value = centrality measure

Getting to Know Data

- Which one will you do first?

1. Find out about average and standard deviation
2. Find out members (range of values)
3. Find out distributions
4. Find out data types
5. Find out missing values
6. Find out interaction between data

Getting to Know Data

- Find out members
- Find range of values?
  - Min? Max?
  - What are unique values

Geo data table

Edit row

Enter filter text

	Location	Gaging station	River	Level	Date	Average maxium water level	Average minimum water level	Water level	Hydrograph
1	Achleiten	Achleiten	Donau	400 cm	2016-06-08 11:30	-	-	Unknown	ganglinien390
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4	Vilshofen	Vilshofen	Donau	413 cm	2016-06-08 10:15	555 cm	299 cm	Normal	ganglinien494
5	Hofkirchen	Hofkirchen	Donau	397 cm	2016-06-08 09:00	557 cm	196 cm	Normal	ganglinien576
6	Deggendorf	Deggendorf	Donau	414 cm	2016-06-08 11:30	615 cm	192 cm	Normal	ganglinien344
7	Pfelling	Pfelling	Donau	497 cm	2016-06-08 09:15	697 cm	268 cm	Normal	ganglinien492
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23	Pretzsch-Mauken	Pretzsch-Mauken	Elbe	165 cm	2016-06-08 11:15	584 cm	71 cm	Normal	ganglinien162
24	Elster	Elster	Elbe	165 cm	2016-06-08 11:15	514 cm	60 cm	Normal	ganglinien140
25	Wittenberg	Wittenberg	Elbe	233 cm	2016-06-08 11:45	543 cm	114 cm	Normal	ganglinien436

Import data | Export data | Map legend

Cartridge	Product Number	Bullet Make	Bullet Type	Bullet Weight Gr	COAL (Inches)
243 Winchester	M6000	Berger	VLD Hunting	90	2.775
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6XC	M6001	Berger	VLD Hunting	115	2.745
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280 Ackley	M6013	Berger	VLD Hunting	168	3.400
7mm RSAUM		Berger	VLD Hunting	168	2.800
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7mm Rem Mag	M6015	Berger	VLD Hunting	168	3.420
7mm Rem Mag	M6016	Berger	VLD Hunting	180	3.420
7mm Wby	M6017	Berger	VLD Hunting	168	3.340
7mm STW	M6018	Berger	VLD Hunting	180	3.670
7mm LRM	M6019	Berger	Hybrid Target	180	3.436
7mm RUM	M6020	Berger	VLD Hunting	180	3.665
28 Nosler	M6041	Berger	Hybrid Target	180	3.610
28 Nosler	M6042	Berger	Elite Hunter	195	3.630
308 Win	M6021	Berger	VLD Hunting	168	2.800
30-06	M6022	Berger	VLD Hunting	168	3.380
300 Win Mag	M6024	Berger	VLD Hunting	190	3.510
300 Wby	M6026	Berger	VLD Hunting	190	3.650
300 RUM		Berger	VLD Hunting	210	3.650
300 RUM		Berger	VLD Hunting	190	3.650
300 RUM		Berger	VLD Hunting	190	3.650
30 Nosler	M6043	Berger	Hybrid Target	215	3.555
300 RUM	M6028	Berger	VLD Hunting	190	3.650
300 RUM	M6035	Berger	Hybrid Target	230	3.650
300 Norma	M6032	Berger	Hybrid Target	230	3.615
338 Norma	M6033	Berger	VLD Hunting	300	3.550
338 Lapua	M6034	Berger	VLD Hunting	300	3.600

Image source: https://www.gunwerks.com/store/featured-products/gunwerks-ammo
https://www.subshell.com/en/sophora/documentation/add-ons/data-driven-journalism-with-maps/index.html

Getting to Know Data

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Getting to Know Data

- Find out distribution
- What is distribution?
  - Frequencies / Probabilities of data
  - A list that shows possibilities of how each member would occur
- Distribution guides us
  - Range of data
  - Candidates for outliers

Histogram

- Specific bar graph representing distribution of data
- x-axis: bins of data values
- y-axis: frequency of data values
- Example:

8 5 4 9 6 5
6 3 8 1 4 5

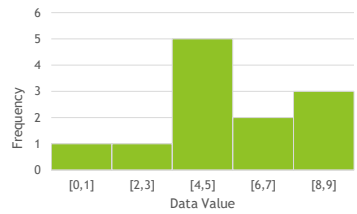
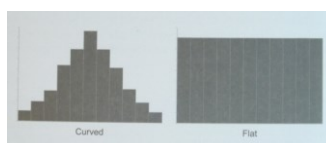
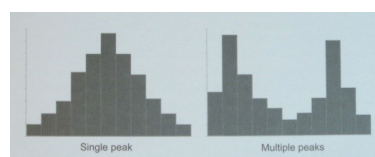


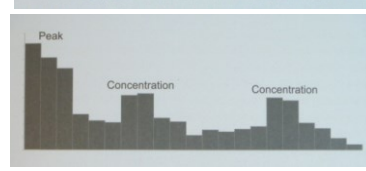
Image source: https://openclipart.org

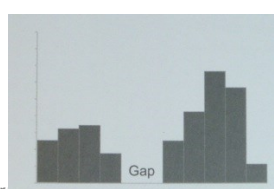
Distribution Patterns

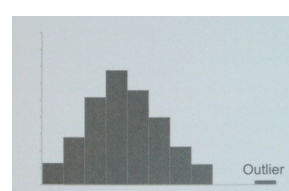


- Curved vs. Flat
- Number of peaks
- Concentration
- Gaps
- Outliers
- Symmetrical vs. Skewed









Source: [2]

Normal or Skewed

- Determine shape of distribution

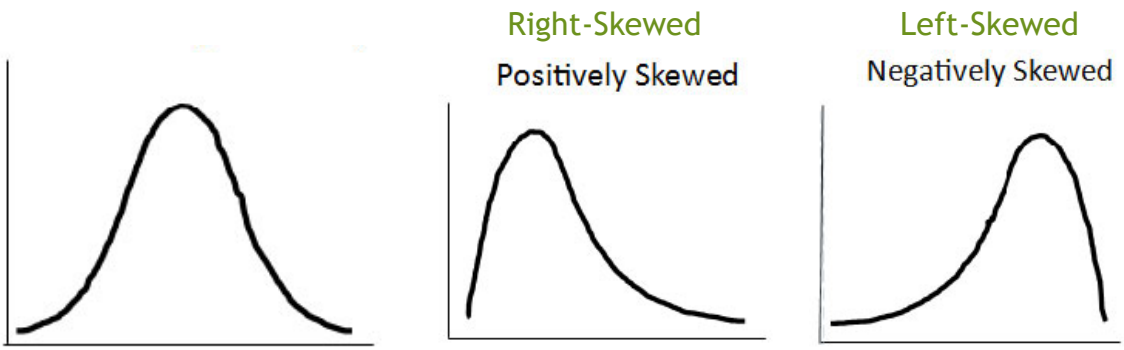
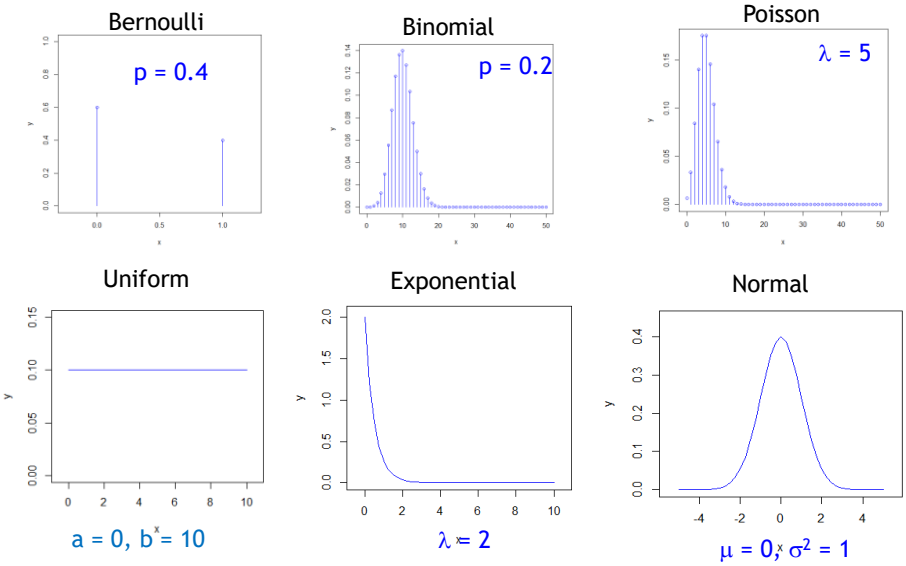


Image source: https://www.aea267.k12.ia.us/system/assets/uploads/images/1558/skew_cuves.jpg
http://www.statisticshowto.com/wp-content/uploads/2014/02/pearson-mode-skewness.jpg

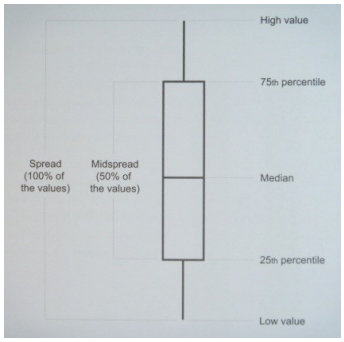
Examples of Distribution Shape

- Well-known distribution functions



Box Plot

- Also call box-and-whisker plot
- Use statistical values to plot range of data
- Does not give frequencies but helps us see spread of data
- Pro: Compare multiple data sets



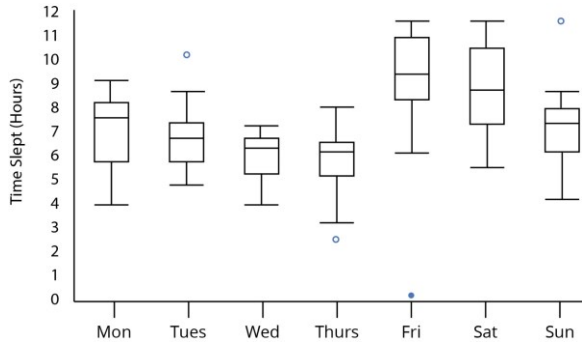


Image source: Figures 10.34 [2]
https://help.plot.ly/what-is-a-box-plot/

Getting to Know Data

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Image source: https://openclipart.org

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Getting to Know Data

- With series of data values, how do we choose “the best” representative of each column?

Geo data table

Edit row

Enter filter text

	Location	Geging station	River	Level	Date	Average maxium water level	Average minimum water level	Water level	Hydrograph
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6	Deggendorf	Deggendorf	Donau	414 cm	2016-06-08 11:30	615 cm	192 cm	Normal	ganglinien344
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Import data | Export data | Map legend

Cartridge	Product Number	Bullet Make	Bullet Type	Bullet Weight Gr	COAL (Inches)
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300 Wby	M6026	Berger	VLD Hunting	190	3.650
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300 RUM		Berger	VLD Hunting	190	3.650
300 RUM		Berger	VLD Hunting	190	3.650
30 Nosler	M6043	Berger	Hybrid Target	215	3.555
300 RUM	M6028	Berger	VLD Hunting	190	3.650
300 RUM	M6035	Berger	Hybrid Target	230	3.650
300 Norma	M6032	Berger	Hybrid Target	230	3.615
338 Norma	M6033	Berger	VLD Hunting	300	3.550
338 Lapua	M6034	Berger	VLD Hunting	300	3.600

Image source: https://www.gunwerks.com/store/featured-products/gunwerks-ammo
https://www.subshell.com/en/sophora/documentation/add-ons/data-driven-journalism-with-maps/index.html

Getting to Know Data

- Measure of Centrality = Representative of data
  - Mean or Average
  - Median
  - Mode

Getting to Know Data

- In our data set, find mean, median, mode of all features?

Provides:	Nominal	Ordinal	Interval	Ratio
The "order" of values is known		✔	✔	✔
"Counts," aka "Frequency of Distribution"	✔	✔	✔	✔
Mode	✔	✔	✔	✔
Median		✔	✔	✔
Mean			✔	✔

Image source: https://www.mymarketresearchmethods.com/types-of-data-nominal-ordinal-interval-ratio/

Locations of mean, median, mode

- Normal vs. Skewed

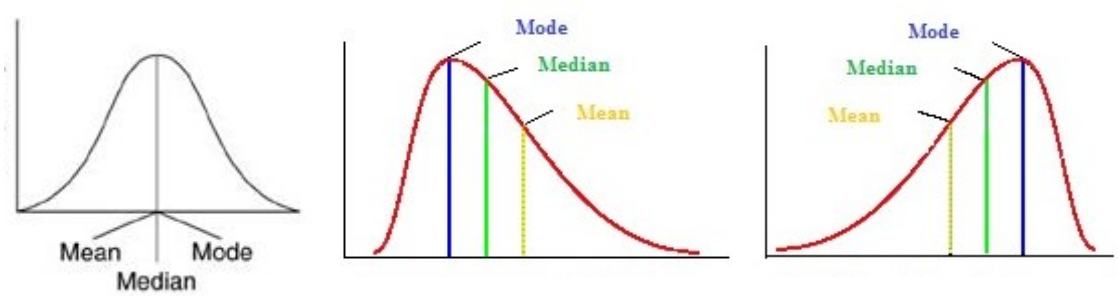

Image source: http://www.statisticshowto.com/wp-content/uploads/2014/02/pearson-mode-skewness.jpg

Time to pick suitable measure of central tendency

- Recommending….

Type of Data	Measure of Central Tendency
Nominal, Categorical	Mode
Ordinal	Median
Interval & Ratio (not skewed)	Mean
Interval & Ratio (skewed)	Median

Source: [2]

Getting to Know Data

- Find out how does each column vary ?
- Find out how much data spread?

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Edit row

Enter filter text

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21	Mühlberg	Mühlberg	Elbe	262 cm	2016-06-08 11:15	684 cm	177 cm	Normal	ganglinien198
22	Torgau	Torgau	Elbe	167 cm	2016-06-08 11:30	623 cm	70 cm	Normal	ganglinien160
23	Pretzsch-Mauken	Pretzsch-Mauken	Elbe	165 cm	2016-06-08 11:15	584 cm	71 cm	Normal	ganglinien162
24	Elster	Elster	Elbe	165 cm	2016-06-08 11:15	514 cm	60 cm	Normal	ganglinien140
25	Wittenberg	Wittenberg	Elbe	233 cm	2016-06-08 11:45	543 cm	114 cm	Normal	ganglinien436

Import data Export data Map legend

Cartridge	Product Number	Bullet Make	Bullet Type	Bullet Weight Gr	COAL (inches)
243 Winchester	M6000	Berger	VLD Hunting	90	2.775
6 Creedmoor		Berger	Hybrid Target	105	2.670
6XC	M6001	Berger	VLD Hunting	115	2.745
25-06 Rem	M6002	Berger	VLD Hunting	115	3.215
6.5 Creedmoor	M6004	Berger	Elite Hunter	140	2.865
6.5-284	M6006	Berger	VLD Hunting	140	3.085
264 Win Mag	M6007	Berger	VLD Hunting	140	3.325
6.5 PRC		Berger	Elite Hunter	140	2.970
280 Remington	M6012	Berger	VLD Hunting	140	3.330
280 Ackley	M6013	Berger	VLD Hunting	168	3.400
7mm RSAUM		Berger	VLD Hunting	168	2.800
7mm RSAUM	M6042	Berger	Hybrid Target	180	2.820
7mm WSM	M6014	Berger	VLD Hunting	168	2.930
7mm Rem Mag	M6015	Berger	VLD Hunting	168	3.420
7mm Rem Mag	M6016	Berger	VLD Hunting	180	3.420
7mm Wby	M6017	Berger	VLD Hunting	168	3.340
7mm STW	M6018	Berger	VLD Hunting	180	3.670
7mm LRM	M6019	Berger	Hybrid Target	180	3.436
7mm RUM	M6020	Berger	VLD Hunting	180	3.665
28 Nosler	M6041	Berger	Hybrid Target	180	3.610
28 Nosler	M6042	Berger	Elite Hunter	195	3.630
308 Win	M6021	Berger	VLD Hunting	168	2.800
30-06	M6022	Berger	VLD Hunting	168	3.380
300 Win Mag	M6024	Berger	VLD Hunting	190	3.510
300 Wby	M6026	Berger	VLD Hunting	190	3.650
300 RUM		Berger	VLD Hunting	210	3.650
300 RUM		Berger	VLD Hunting	190	3.650
300 RUM		Berger	VLD Hunting	190	3.650
30 Nosler	M6043	Berger	Hybrid Target	215	3.555
300 RUM	M6028	Berger	VLD Hunting	190	3.650
300 RUM	M6035	Berger	Hybrid Target	230	3.650
300 Norma	M6032	Berger	Hybrid Target	230	3.615
338 Norma	M6033	Berger	VLD Hunting	300	3.550
338 Lapua	M6034	Berger	VLD Hunting	300	3.600

Image source: https://www.gunwerks.com/store/featured-products/gunwerks-ammo
https://www.subshell.com/en/sophora/documentation/add-ons/data-driven-journalism-with-maps/index.html

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Variance

- Squared value of standard deviation
- Computed by

$$s^2 = \frac{\sum_{i=1}^{N}(x_i - \bar{x})^2}{N-1}$$

$\bar{x} = Average$
$x_i = Each\ data$
$N = Number\ of\ data$

- Example:

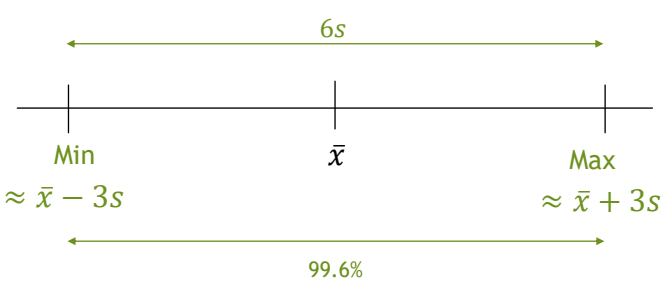
8 2 4 9 7 $\quad \bar{x} = 6.0$

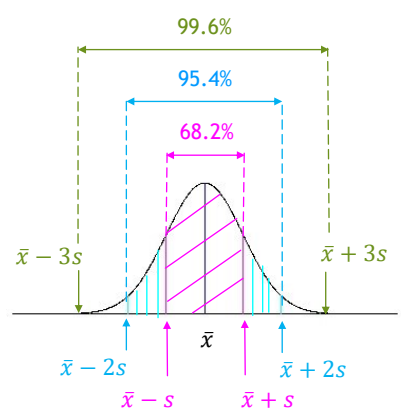
$$s^2 = \frac{(8-6)^2+(2-6)^2+(4-6)^2+(9-6)^2+(7-6)^2}{5-1} = 8.53$$

Image source: https://openclipart.org

Standard Deviation

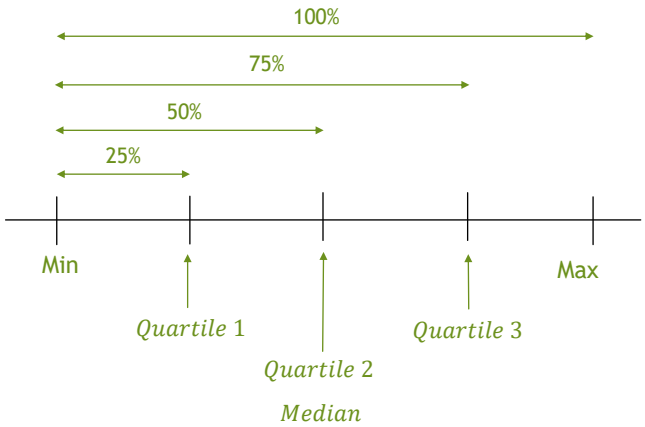
- What does standard deviation tell us?
- In general, between min and max, <u>data spread out about $6s$</u>.

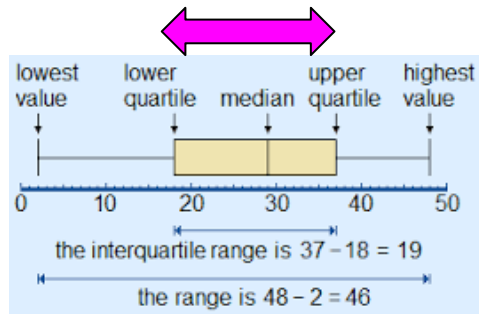




Quartile and Inter-quartile Range (IQR)

- Values that split data into 4 equal partitions





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Outliers

- Outliers can be determined from IQR

If data value > Q3 + 1.5IQR or
data value < Q1 - 1.5IQR,
we consider such value to be outlier.

If data are normal &
data value > Mean + 3SD or
data value < Mean – 3SD
we consider such value to be outlier.

If data value > 95% percentile or
data value < 5% percentile
we consider such value to be outlier.

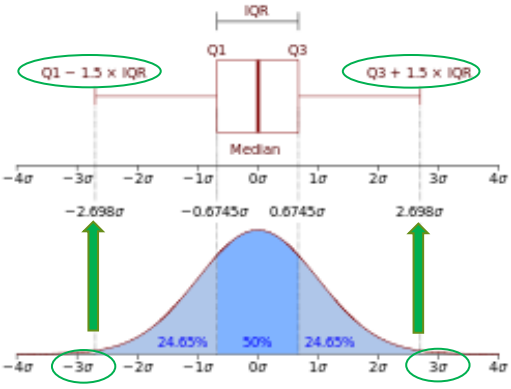

Image source: https://upload.wikimedia.org/wikipedia/commons/thumb/1/1a/Boxplot_vs_PDF.svg/250px-Boxplot_vs_PDF.svg.png

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Getting to Know Data

- Which one will you do first?

1. Find out about average and standard deviation
2. Find out members (range of values)
3. Find out distributions
4. Find out data types
5. Find out missing values
6. Find out interaction between data

Bivariate data analysis

Image source: https://openclipart.org

Source: Data Exploration and Visualization, Kitsana Waiyamai

Example

- Each data would be more meaningful if we study with others How can we "see" relationship between each data

Image source: https://openclipart.org

Correlation

- Correlation coefficient used as numerical value to measure correlation
- Paired data $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$

$$\rho_{X,Y} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

- Range of $\rho_{X,Y}$: $-1 \le \rho_{X,Y} \le 1$
- If $\rho_{X,Y} = 0$, then X and Y are independent.

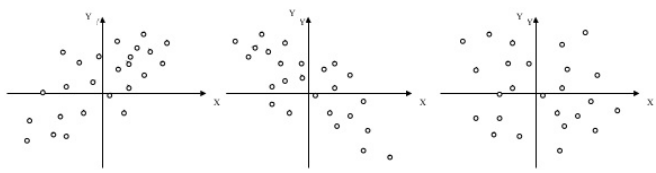

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Image source: http://www.slideshare.net/AhmedShahid/t-tests-anovas-and-regression

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Correlation

- Strength
- Negative correlations are similar

[0.6,0.8)
Very strong
[0.8,1.0]

[0.2,0.4)
Moderate
[0.4,0.6)

[0.0,0.2)

Source: http://www.statstutor.ac.uk/resources/uploaded/pearsons.pdf
Image source: https://support.minitab.com/en-us/minitab-express/1/help-and-how-to/graphs/scatterplot/create-the-graph/choose-a-scatterplot/

Bubble Plot

- Example 4: >=3 variables
- Often use bubble scatter plot (4 variables) or scatter plot matrix

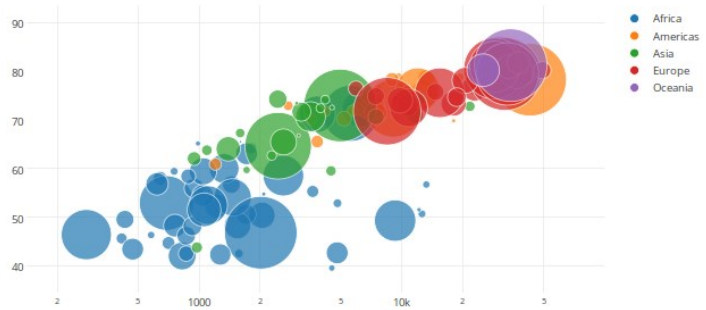
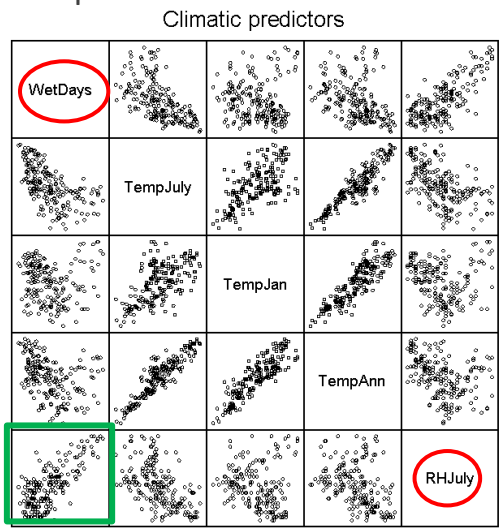


Image source: http://i.imgur.com/ggQMO8z.png

Source: [1]

Scatterplot Matrix

- Example:



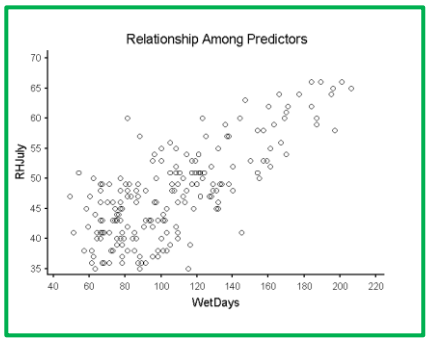
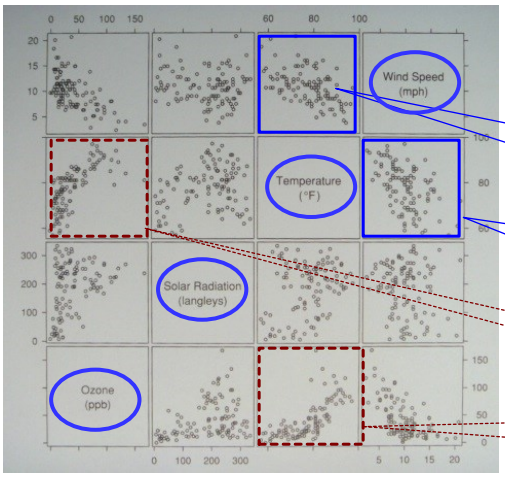


Image source:
https://www.pcord.com/nscatterplot.htm
https://www.pcord.com/nscatterplotmatrix.htm

Scatterplot Matrix (cont.)

- Display correlation of **multiple pairs** of variables in the same time.

4 variables:
- Wind Speed
- Temperature
- Solar Radiation
- Ozone



X = temperature, Y = wind speed

X = wind speed, Y = temperature

Same pair of variables

X = ozone, Y = temperature

X = temperature, Y= ozone

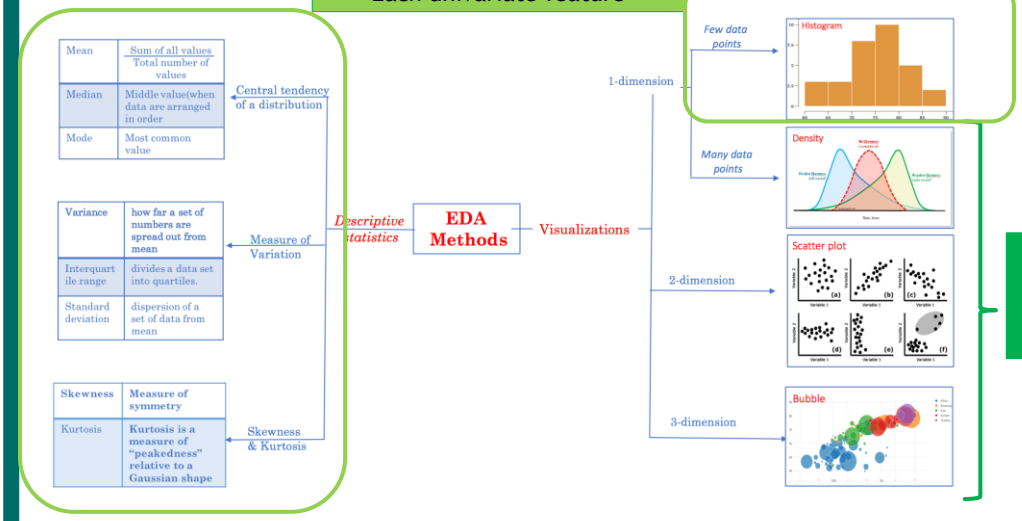
Same pair of variables

Source: Figure 11.22, [1]

Only need half of scatterplot matrix

Exploratory Data Analysis : Steps

Analyze Each univariate feature



Detect interactions among features

Image Source: https://www.netsolutions.com/insights/data-visualization-a-building-block-of-an-intelligent-enterprise/

Conclusion

- Data exploration allows us to find basic information about the data
  - Find out data types
  - Handle missing values
  - Numerical data
    - Find out distributions or patterns
    - Find out statistical values : centrality and dispersion
    - Find out correlation
  - Categorical data
    - Find out frequencies and probabilities